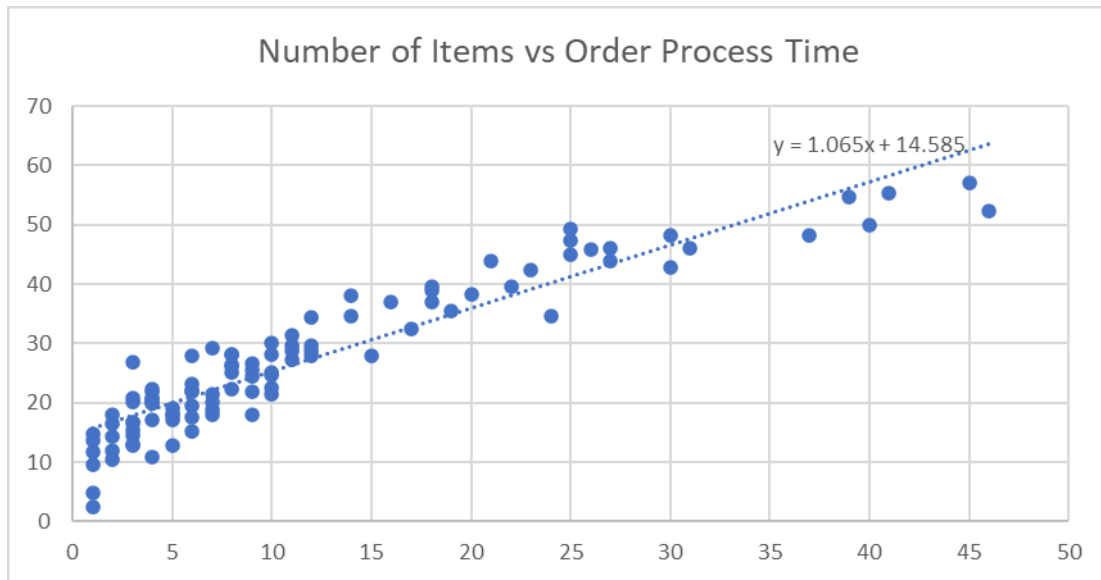


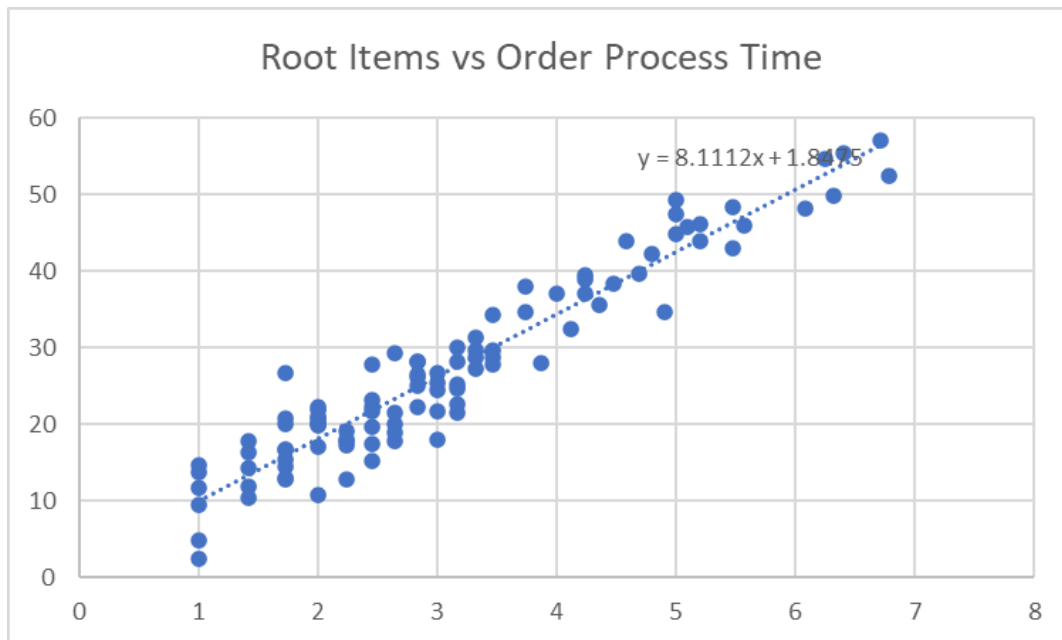
8

a) and b)



B0 and B1 calculated on excel

c)



B0 and B1 calculated on excel

d) The model using the square root of the items seems to have a better fit for the process time

e)

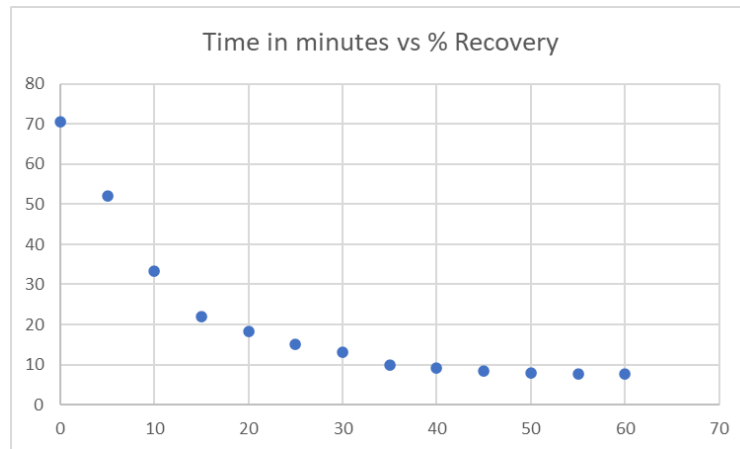
Original model: $1.065(13) + 14.585 = 28.23$

Square root model: $8.1112(\sqrt{13}) + 1.847 = 31.09$

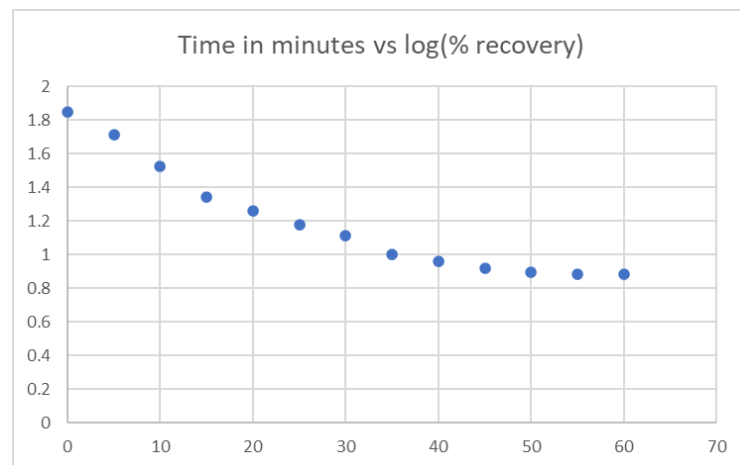
These values are fairly similar.

18. Tables created using excel

a)



b)



19 (for log model)

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	1.15275	1.15275	93.08	<.0001
Error	11	0.13624	0.01239		
Corrected Total	12	1.28899			

Root MSE	0.11128	R-Square	0.8943
Dependent Mean	1.19507	Adj R-Sq	0.8847
Coeff Var	9.31229		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	1.67258	0.05833	28.67	<.0001
time	1	-0.01592	0.00185	-8.55	<.0001

a) $y = 1.67258 - 0.01592x + \epsilon$

b) $\sigma_E = \frac{MSE}{n-2} = 0.01239$

c) $se(\beta_0) = 0.05833$

$se(\beta_1) = 0.00165$

30.

$$\hat{y} \pm t_{0.025,11} s_{\varepsilon} \sqrt{\frac{1}{n} + \frac{(30-x)^2}{S_{xx}}} = 1.195 \pm (2.201)(0.1114)\sqrt{1/13} \rightarrow (1.127, 1.263)$$

31.

a)

$$\hat{y} \pm t_{0.025,11} s_{\varepsilon} \sqrt{1 + \frac{1}{n} + \frac{(30-x)^2}{S_{xx}}} = 1.195 \pm (2.201)(0.1114)\sqrt{1 + 1/13} \rightarrow (0.941, 1.449)$$

b) The above confidence interval is wider than the confidence interval on the mean.

c) The observed value has more variance than the mean, so its confidence interval is wider at the same level of significance. That is, it is easier to predict the average value of the data than it is to predict a single observation, hence the mean has less variability.

45.

$$\rho_{xy} = \frac{cov(x,y)}{\sqrt{Var(x)Var(y)}}$$

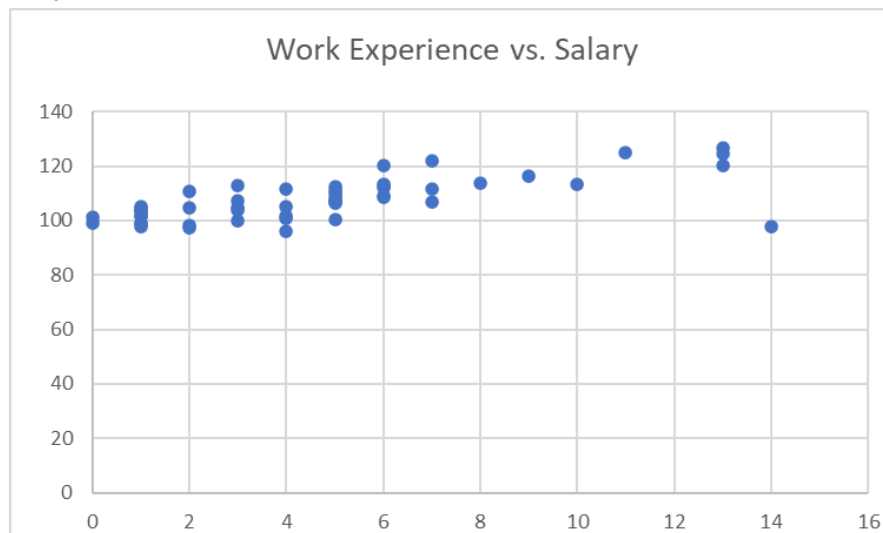
$$cov(x,y) = \Sigma(x_i - \bar{x})(y_i - \bar{y})/n = 960.4462/53 = 18.16123$$

$$Var(x) = 12.632$$

$$Var(y) = 58.484$$

$$\rho_{xy} = \frac{18.16123}{\sqrt{(12.632)(58.48)}} = 0.692863 \text{ Positive correlation does appear to agree with the scatter plot.}$$

The value is not particularly close to 1, which suggests work experience and salary are not perfectly positively correlated, which follows the data and scatter plot



46. Done in SAS

$$a) \hat{\beta}_0 = 100.6157, \hat{\beta}_1 = 1.49062$$

$$b) \sigma_E = \frac{MSE}{n-2} = 30.8686$$

- c) There does seem to be a significant relationship between salary and experience since the p-value is less than 0.001, making it significant at any reasonable α level
- d) $R^2 = 0.4825$, 48.25% of the variability in salary is explained by years of experience

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	1439.26948	1439.26948	46.63	<.0001
Error	50	1543.42976	30.86860		
Corrected Total	51	2982.69923			

Root MSE	5.55695	R-Square	0.4825
Dependent Mean	107.69815	Adj R-Sq	0.4722
Coeff Var	5.15891		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	100.61570	1.29184	77.89	<.0001
exper	1	1.49062	0.21930	6.83	<.0001